MIL-H-52036D
7 September 1976
SUPERSEDING
MIL-H-52036C
22 December 1967

MILITARY SPECIFICATION

HEATER, WATER: OIL-FIRED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an oil-fired, automatically-controlled, water heater capable of raising the temperature of water from 32° F to 70° F at the rate of 3,000 gallons of water per hour without contaminating or otherwise affecting the potability.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81	- Tags, Shipping and Stock.
VV-F-800	- Fuel Oil, Diesel.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

FSC 4410

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MIL-C-104	- Crates, Wood; Lumber and Plywood. Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronics, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

FEDERAL

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Tests: Exponential Distribution.

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DRAWINGS

ME

TA13213E5980 - Heater, Water, Oil Fired.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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3. REQUIREMENTS

3.1 Description. The water heater shall be as shown on Top Assembly TA13213E5980 and as specified herein.

3.1.1 Operating characteristics. The water heater shall produce a temperature rise of not less than 38° F in water flowing through the water heater at the rate of 3000 gph without contaminating or otherwise affecting the potability. The heater shall automatically start when the effluent water temperature is less than 70° F, shall automatically shutdown when the effluent water temperature exceeds 75° F and shall provide flame failure, low water, and high water temperature protection.

3.1.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the water heater. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one water heater for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce water heaters that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more water heaters for inspection as specified in 4.4.

3.4 Material. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. The material selected shall have no adverse effect on health

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of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate medical service who will act as an advisor to the procuring activity (see 6.5).

3.5 Cable assembly. A cable assemble as shown on Figures 1 and 2 shall be furnished with each water heater.

3.6 Reliability. The specified Mean-Time-Between Failure (MTBF) shall be 60 hours when the water heater is tested as specified in 4.6.2.1.

3.7 Maintenance ratio. The water heater shall have a maintenance ratio of not more than 0.025. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair or replaced components and scheduled before-and-after operation shall be excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.8 Electromagnetic interference. The electromagnetic interference emission and susceptability of the water heater shall conform to MIL-STD-461.

3.9 Lubrication.

3.9.1 Lubricants. The water heater shall operate as specified herein when serviced with the following military lubricants, grades as applicable to the operating conditions:

Grease lubrication: MIL-G-10924 or MIL-G-23827.

3.9.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, Types I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extention. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.9.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

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3.9.4 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

(a) MIL-G-23827 - For ball, roller, and needle bearings, gears, and on sliding and rolling surfaces of such equipment as mechanical control systems; for use on gears, actuator screws and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of minus 100° F to plus 250° F.
(b) MIL-G-10924 - For lubrication and surface corrosion protection of automotive and artillery equipment operated over the temperature range of minus 65° F to plus 225° F. Also used in other applications within this temperature range where grease with oxidation-resistant and corrosion-preventive properties is desirable. Not for use on machinery which comes in contact with foods or potable water.

The water heater shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.10 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.10.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

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3.10.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

3.11 Identification marking. The water heater shall be identified in accordance with MIL-STD-130. The marking shall be applied to the water heater on identification plates conforming to MIL-P-514, Type I, Style 1, Composition C, Grade A, Class 1. Plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.12 Workmanship.

3.12.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.12.2 Welding.

3.12.2.1 Aluminum welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or spatter.

3.12.2.2 Steel welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials,

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weld filler metals and fabrication techniques shall be as required to enable the water heater to conform to the examination specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the water heater shall be at the option of the contractor.

3.12.3 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

3.12.4 Welding practices.

(a) Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the contractor's recommendations.
(b) Tack welds shall be subject to the same quality requirements as final welds.
(c) Work shall be positioned for flat welding whenever practicable.
(d) Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part or the water heater.

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3.12.5 Weld quality.

(a) The contractor shall be responsible for determining the inspections needed to insure that, when the welded parts are assembled together to make the water heater, the water heater shall conform to the inspection requirements specified in Section 4.
(b) All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.
(c) Undercut in weldments shall be held to a minimum and shall not be more than 1/32 inch deep.

3.12.6 Brazing. Brazing of steels, copper alloys and nickel alloys shall conform to MIL-B-7883.

3.12.7 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

3.12.8 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.12.9 Cleaning. The water heater shall be cleaned of all cutting oil, residue of tests, grease, dirt, metal scrap, and other contaminants. Cleaning shall be accomplished in a manner that will not leave a residue or otherwise make the water heater unsuitable for its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specification, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any test or examination by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.9 and 6.10).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The water heater shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The water heater shall be tested as specified in 4.6.2.1 through 4.6.2.2. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production water heaters will be selected at random by the Government from the water heaters being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in

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4.6.2.1 through 4.6.2.5 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production water heater shall not exclude the remaining water heaters from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production water heater to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the initial production water heater and shall be cause for refusal by the Government to continue acceptance of production water heaters until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at not cost to the Government on water heaters previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all water heaters accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined as specified in 4.6.1. AQL shall be 2.5 percent defective for major defects and 4.0 defective for minor defects.

4.5.3 Tests. Each water heater shall be tested as specified in 4.6.2.3 through 4.6.2.3.4. Failure of the test shall be cause for rejection.

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4.6 Inspection procedure.

4.6.1 Examination. The water heater shall be examined for the following defects:

Major
101. Dimensions not as shown on the drawings.
102. Assembly not complete or not correct.
103. Parts or components missing.
104. Parts or components do not function.
105. Material not as specified.

Minor
201. Treatment and painting not as specified.
202. Lubrication devices not accessible.
203. Workmanship not as specified.
204. Heater not completely drained and cleaned before packaging.
205. Identification marking not as specified.

4.6.2 Tests.

4.6.2.1 Operational tests (6 hours). The operational tests shall be conducted using the fuel specified herein and tap water. The tap water shall be at a temperature below 70° F. The water heater shall be started and operated for 6 hours. Fuel shall conform to VV-F-800, Grade DF-2.

4.6.2.1.1 Operation. During the 6 hours of operation, the water heater shall be checked for flue gas and fuel oil leaks, and the temperature of the effluent water shall be noted. Any flue gas or fuel oil leaks shall constitute failure. Inability of the water heater to shut down automatically when the effluent water temperature is more than 75° F or inability to restart when the effluent water temperature falls below 70° F shall constitute failure of this test.

4.6.2.1.2 Flame failure. While the water heater is in operation, simulate flame failure and note the operation of the photoelectric cell, the combustion controller, and the programmer. Inability of the photoelectric cell to actuate the combustion controller; inability of the combustion controller to provide power to the flame failure indicator light and the programmer during starting; inability of the controller to shut down the heater after an absence of flame of 35 to 55 seconds duration during restarting; inability of the controller to revert to

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its starting position within 0.8 seconds after initial flame failure and inability of the controller to attempt to restart the heater; inability of the programmer to energize the transformer to provide ignition arc immediately when provided with power from the combustion controller; inability of the programmer to lock after ignition; or inability of the programmer to reset to normal starting position when power from the controller is stopped, shall constitute failure of this test. At the completion of this phase of operational tests, the combustion controller shall be allowed to cool for 5 minutes before restarting.

4.6.2.1.3 Low-water relay. While the water heater is in operation, shut off incoming water, open the drain valve on the water heater, and note the operation of the low-water relay. Inability of the low-water relay to shut down the water heater when the water falls below the probe of the low-water electrode shall constitute failure of this test.

4.6.2.1.4 High-limit temperature control. Raise the temperature of the water in the water heater until the high-limit temperature control actuates. This can be accomplished by resetting or bypassing the effluent water temperature control and restricting the flow of water through the heater. Inability of the control to shut down the heater when the temperature exceeds 180° F shall constitute failure of this test.

4.6.2.1.5 Capacity. During the final 2 hours of the operational test, water shall be supplied to the heater at a rate of not less than 3000 gph or more than 3050 gph. The influent and effluent temperatures shall be recorded at 15 minute intervals for the 2 hours. A temperature rise of less than 38° F shall constitute failure of this test. To permit the 38° F temperature rise, influent water may be cooled or the effluent water temperature control may be reset or bypassed.

4.6.2.2 Electromagnetic interference. The preproduction model, equipped for the reduction of electromagnetic interference in accordance with 3.8, shall be tested as specified in MIL-STD-461. The contractor shall furnish the contracting officer the test report required by MIL-STD-461, at the time interval specified (see 6.2). Upon approval of the report by the contracting officer and provided all other requirements of the specification are met, the preproduction model shall be used as a model for all production heaters. Disapproval of the report shall constitute failure of this test.

4.6.2.3 Operational test (three-cycle). The water heater shall be operated so as to perform not less than three on-off cycles of effluent water temperature control. Inability of the heater to shut down automatically

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four plies of thickness with no cut edges exposed. The fabric ... be a minimum of

when effluent water temperature is more than 75° F or inability to restart when the effluent water temperature falls below 70° F shall constitute failure of this test. Any flue gas or fuel oil leaks shall also constitute failure of this test.

4.6.2.3.1 Flame failure. Flame failure shall be determined in accordance with 4.6.2.1.2.

4.6.2.3.2 Low-water relay. Low-water relay shall be determined in accordance with 4.6.2.1.3.

4.6.2.3.3 High-limit temperature control. High-limit temperature control shall be determined in accordance with 4.6.2.1.4.

4.6.2.3.4 Capacity. The water heater shall be operated with water supplied to the heater at a rate of not less than 3000 gph nor more than 3050 gph. A water temperature rise of less than 38° F shall constitute failure of this test. To permit the 38° F temperature rise, influent water may be cooled or the effluent water temperature control may be reset or bypassed.

4.6.2.4 Reliability. The reliability specified in 3.6 shall be demonstrated by conducting a reliability test. The test profile shall be as specified in 4.6.2.4.1. Failures shall be assigned in accordance with 4.6.2.4.2. Testing shall be conducted in accordance with Test Plan IV of MIL-STD-781. Testing shall continue until either an "accept" or "reject" decision is reached.

4.6.2.4.1 Reliability test profile. The reliability test shall be conducted for the period specified in 4.6.2.4. During each 6 hours of operation the heater shall be operated according to the following profile. The influent water temperature shall be as specified in 4.6.2.1. The capacity criteria specified in 4.6.2.1.5 shall apply for each 6 hour period. Failures shall be assigned in accordance with 4.6.2.4.2.

4.6.2.4.2 Failure definition. A failure for the purpose of this test is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, degradation below the operating characteristics as required by this specification or causes a critical safety hazard to personnel or equipment which cannot be corrected within 1 hour using only controls, tools, and parts available at the organizational level of maintenance.

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4.6.2.5 Maintenance support evaluation. When specified (see 6.2), maintenance operation testing shall be accomplished to determine conformance to 3.7. Failure of the support elements to maintain the water heater as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.7 Inspection comparison. The Government may select water heaters at any time during the contract production period and subject these water heaters to the examination specified in 4.6.1 and the tests specified in 4.6.2.1 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on water heaters selected at random from those which have been accepted by the Government and will not include the previosuly inspected preproduction model and initial production water heater. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison water heater to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison water heater and shall be cause for refusal by the Government to continue acceptance of production water heaters until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on water heaters previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all water heaters accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective for major defects and 4.0 defective for minor defects.

Major	
106.	Materials, methods, and containers not as specified for Level A or B. Each incorrect material, method, or container shall be considered one defect.
107.	Preservatives applied to surfaces which will come in contact with potable water for Level A, B, or Commercial.
108.	Depreservation guide not prepared and attached as specified for Level A or B, or Commercial.
109.	Disassembled components not matchmarked as specified for Level A or B.
110.	Electric motors not preserved in accordance with the referenced document for Level A or B.
111.	Openings into electrical components and exposed ends of wire, not sealed with tape as specified for Level A.
112.	Belts not released from tension or removed as specified for Level A or B.
113.	Gaskets not secured as specified for Level A or B.
114.	Fuel supply line not coiled, secured, and openings sealed as specified for Level A or B.
115.	Miscellaneous equipment requiring protection not preserved as specified for Level A or B.
116.	Repair parts not preserved in accordance with the referenced document for Level A or B.
Minor	
206.	Consolidation not as specified for Level A or B.
207.	Packing not as specified for Level A.
208.	Marking missing, illegible, incorrect, or incomplete for Level A or B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or B, or Commercial as specified (see 6.2).

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5.1.1 Caution. Contact preservatives shall not be applied to any surface which will come in contact with potable water. The water heater shall be completely drained and the water drain valves shall be left open.

5.1.2 Depreservation guide. A depreservation guide shall be prepared. Place the guide in a waterproof envelope marked "Depreservation Guide" and secure in a conspicuous location on the heater. Unless otherwise specified (see 6.2 and 6.6), DA Form 2258 - Depreservation Guide for Vehicles and Equipment shall be used.

5.1.3 Level A.

5.1.3.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts vulnerable to damage or loss and to accomplish the packing specified herein. Specifically the photoelectric cell assembly, stack assembly, roof hood assembly, air duct assembly, outlet piping, lever coupling assembly, fuel supply line, and flue retarders shall be removed from the heater. Bolts, nuts, pins, screws, and washers involved shall be reinstalled in one of the mating parts and secured to prevent their loss.

5.1.3.2 Matchmarking. To facilitate reassembly, removed parts and mating parts shall be matchmarked. Matchmarking information shall be on cloth tags conforming to UU-T-81, Type A. The information on the tags shall be applied with waterproof ink. The tags shall be securely attached to the parts so as to prevent damage to the preservation or to the parts.

5.1.3.3 Preservatives. Surfaces of components and parts of the water heater requiring the application of a contact preservative in accordance with MIL-P-116, except as specified in 5.1.1, shall be coated with preservatives conforming to the applicable specifications listed in and shall be applied in accordance with MIL-P-116. Surfaces of bolts, nuts, linkages, pins, grease fittings, flue retarders, levers, wire rope, clips, clamps, cap screws, and couplings or any other surfaces requiring the application of a contact preservative not definitely specified herein shall be coated with Type P-1 preservative.

5.1.3.4 Electrical components.

5.1.3.4.1 Photoelectric cell assembly. The photoelectric cell assembly shall be preserved in accordance with MIL-P-116, Method IA-14. The inner container shall be the contractor's standard commercial container or a

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close-fitting box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, grade as applicable. The outer container shall conform to PPP-B-636, Type CF, Class Weather-Resistant, Variety SW, grade as applicable.

5.1.3.4.2 Electric motors. Electric motors shall be preserved in accordance with MIL-E-16298, by the alternate method specified therein

5.1.3.4.3 Control panel boxes, low-water cut-out electrode assembly, switches, solenoids, starter boxes and transformers. All openings into these items shall be sealed with tape conforming to PPP-T-60, Type IV.

5.1.3.4.4 Electric wiring system. Exposed ends of wires, plug openings, sockets, connectors, terminals, receptacles, and openings of conduit and junction boxes shall be sealed with tape conforming to PPP-T-60, Type IV.

5.1.3.4.5 Cable assemblies. The cable assemblies shall be individually coiled to not less than 10 times their diameter and the coils secured with equally spaced wraps of cotton tape or twine or tape conforming to PPP-T-60, Type IV.

5.1.3.5 Belts and pulleys. Belts shall be removed or released from tension. Unpainted pulleys shall be coated with primer conforming to TT-P-664. Removed belts shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.3.6 Gaskets. Gaskets exposed by disassembly shall be secured in place, on the fixed component, with tape conforming to PPP-T-60, Type IV.

5.1.3.7 Fuel supply line. The fuel supply line shall be coiled and the coil secured with cotton tape or twine or with tape conforming to PPP-T-60, Type IV. Openings shall be sealed with plastic or metal caps or plugs as applicable or with tape specified herein.

5.1.3.8 Flue retarders. Flue retarders shall be preserved in accordance with MIL-P-116, Method III, applying Type P-1 preservative, and using a close-fitting box conforming to PPP-B-636, W5c or W6c as applicable.

5.1.3.9 Miscellaneous equipment. Components of equipment not specifically identified herein which require protection from corrosion and mechanical damage shall be preserved as specified herein for components of similar design and construction.

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5.1.3.10 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.1.3.11 Technical publications. Technical publications for each heater shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.3.12 Maintenance tools and operating tools. Maintenance tools and operating tools shall be preserved in accordance with Level A preservation and packaging requirements of PPP-P-40.

5.1.3.13 Consolidation.

5.1.3.13.1 Large components. Large components removed from the water heater such as the stack assembly, roof hood assembly, air duct assembly, outlet piping assembly, and fuel supply line shall be secured to the water heater or base in a manner to prevent damage to the items and equipment and to keep the overall cube at a minimum. Securing shall be by bolting or by strapping. Strapping shall conform to QQ-S-781, Class 1, Type I or II, Finish B, Grade 2, size as applicable.

5.1.3.13.2 Small components. Small preserved and unpreserved components such as repair parts, tools, publications, photoelectric cell assembly, flue retarders, removed belts, and any other small components not secured to the water heater shall be consolidated together in close-fitting boxes conforming to PPP-B-636, W5c or W6c as applicable. Cushioning and blocking shall be provided to prevent damage to the contents. Strapping of boxes shall not be required. The size of the boxes shall be determined by the available space for securing the boxes to the base or to the heater without increasing the overall cube. Securing to the base or to the heater shall be as specified in 5.1.3.13.1.

5.1.4 Level B. Each complete water heater shall be preserved as specified for Level A with the following exceptions.

5.1.4.1 Photoelectric cell assembly. The photoelectric cell assembly shall be preserved in accordance with MIL-P-116, Method IC-2 using the contractor's standard commercial container or a close-fitting box conforming to PPP-B-636, Type CF, Class Domestic, Variety SW, grade as applicable.

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5.1.4.2 Control panel boxes, low-water cut-out electrode assembly, switches, solenoids, starter boxes and transformers. Sealing of openings shall not be required.

5.1.4.3 Electric wiring system. Sealing of openings and exposed ends of wire with tape shall not be required.

5.1.4.4 Belts and pulleys. Belts shall be removed or released from tension. Removed belts shall be secured to the heater in a protected location near the place of removal.

5.1.4.5 Flue retarders. The retarders shall be preserved in accordance with MIL-P-116, Method III, using a container as specified in 5.1.4.1.

5.1.5 Commercial. Each complete heater shall be preserved in accordance with FED. STD. No. 356.

5.2 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A. The water heater complete with all components shall be enclosed within a housing. The housing shall consist of a top, two sides, and two ends constructed and assembled in accordance with MIL-C-104, Type II, Class 2. The sheathing shall be cut to allow sides to fit over lifting flanges of the base, and the end and side sheathing shall extend to the lower edge of the wood members attached to the base. The housing shall be secured to the wood members on the sides and ends of the base in accordance with the crate specification as specified for Type II crates.

5.2.2 Commercial. Each complete heater shall be packed in accordance with FED. STD. No. 356.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2. Commercial. Marking for commercial packaging shall be in accordance with FED. STD. No. 356.

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6. NOTES

6.1 Intended use. The water heater covered by this specification is intended for heating potable water under arctic conditions before the water is stored in tanks.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examinations and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2, 4.6.2.4, and 4.6.2.5).
(d) When initial production inspection is required and number of water heaters to be furnished when applicable (see 3.3 and 4.4).
(e) When the welding process will not be at the option of the contractor (see 3.12.2.2).
(f) Time interval required for submission of electromagnetic interference report (see 4.6.2.2 and 6.4).
(g) Degree of preservation and packing required (see 5.1 and 5.2).
(h) When other than DA Form 2258 is to be used (see 5.1.2 and 6.7).

6.3 Preproduction model. Any changes or deviations of production water heaters from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish water heaters conforming to this specification.

6.4 Electromagnetic interference. The contracting officer should forward a copy of the test report (see 4.6.2.2) to the Commander, USA Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, NJ 07703.

6.5 For Army procurement, the Surgeon General of the Army will act as advisor to make the necessary determination on health hazards.

6.6 DA Form 2258. The contracting officer should arrange to furnish DA Form 2258 when requested by the contractor.

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6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each water heater.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of water heaters.

6.9 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken, when applicable."

6.10 Definitions.

6.10.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractural requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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Air Force - 99

Review activities:
Army - EL, SM

User activity:
Navy - CG

Preparing activity:
Army - ME

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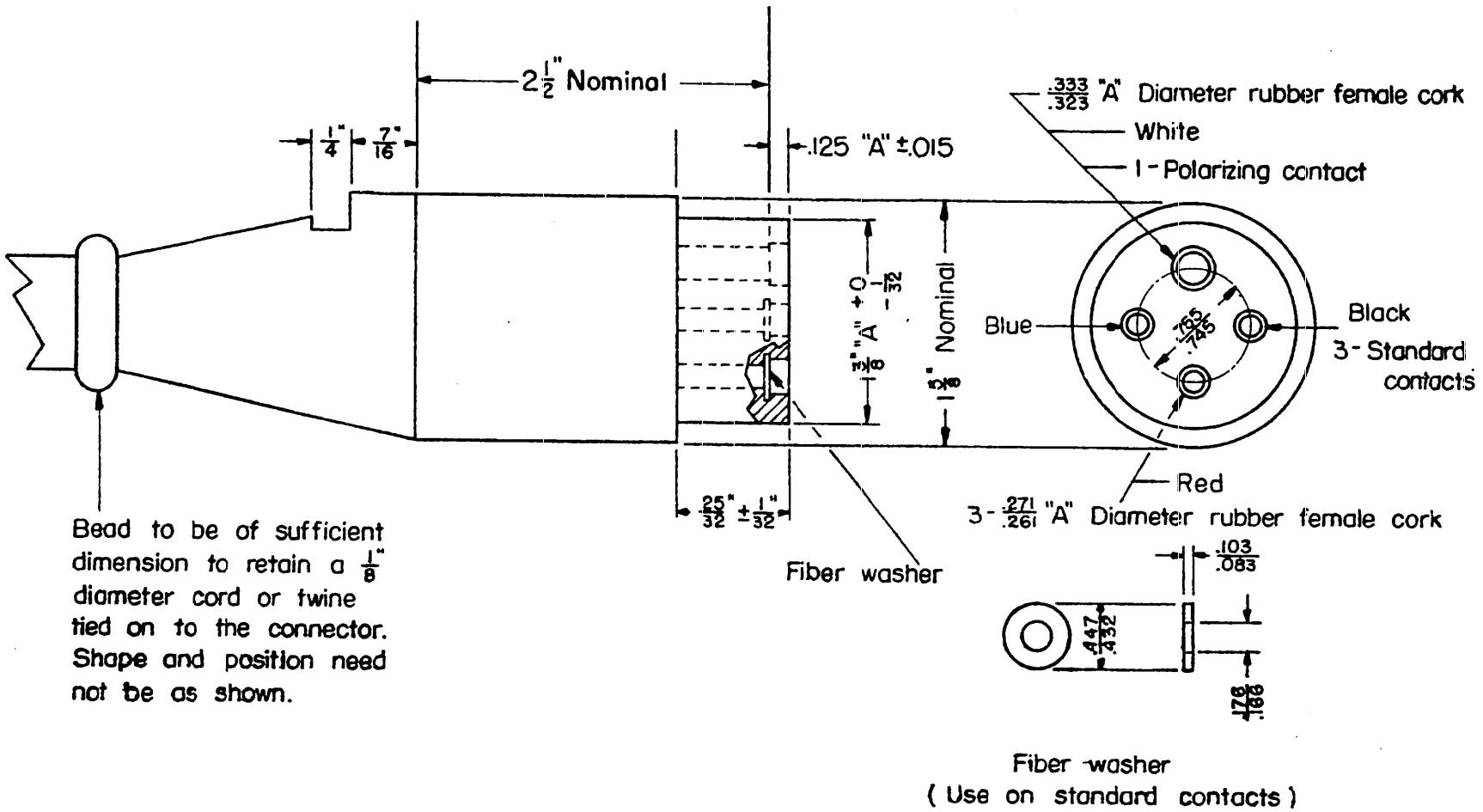


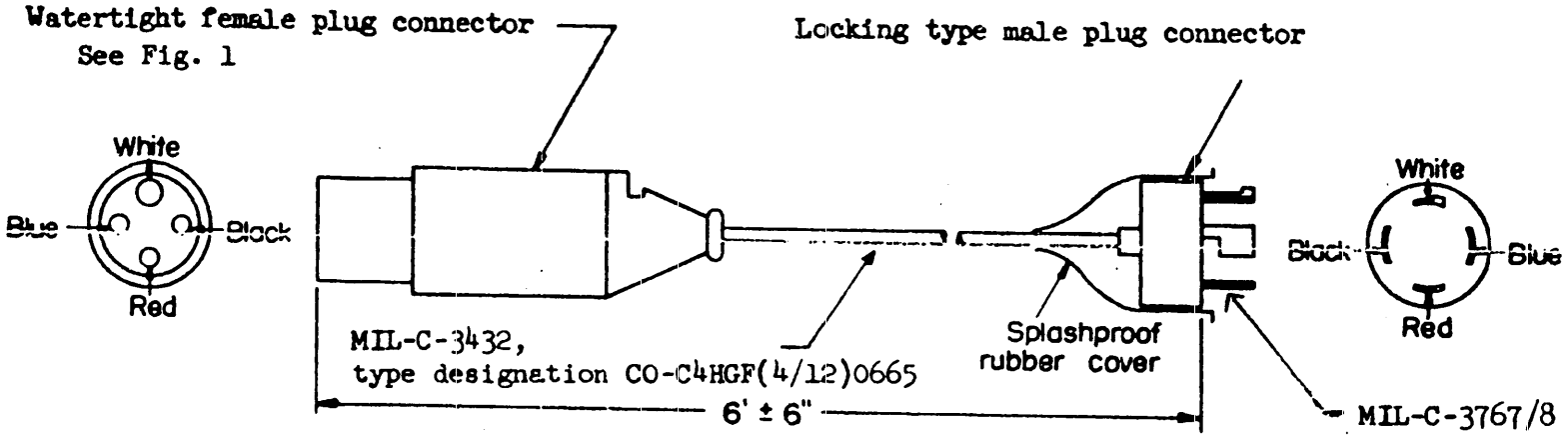
FIGURE 1-WATERTIGHT FEMALE PLUG CONNECTOR

ARMY FORT BELVOIR VA

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Note: Molded lock type cap acceptable.
If molded cap is used, a $\frac{1}{4}$" skirt shall be provided for splashproofness.
Color coding shall be as indicated.

FIGURE 2 – CABLE ASSEMBLY, POWER, ELECTRICAL, 4-CONDUCTOR, ONE END W/WATERTIGHT FEMALE PLUG CONNECTOR, OTHER END W/LOCKING TYPE MALE PLUG CONNECTOR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-H-52036D Heater, Water: Oil-Fired

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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